

Muhammed Shehzad Hassan

M.Sc. Materials Engineering, RWTH Aachen

Structural simulation | Composites | Failure assessment

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Portfolio shehzadhassanbiz-cell.github.io/shehzadhassan.github.io



Availability

Degree

Master's thesis

Portfolio

Languages

Internship: immediately, full-time Permanent role: from March 2027

M.Sc. Materials Engineering (Structural Integrity), RWTH Aachen, expected Mar 2027

Fraunhofer IPT: Digital Shadow for recycled carbon fibre tapes

Ansys simulation projects, each validated against a hand calculation

English (C1), German (A1/A2, improving)

Profile

Materials engineer (M.Sc. RWTH Aachen, Structural Integrity) focusing on composites, structural simulation and failure analysis. In my Master's thesis at Fraunhofer IPT, I am developing a Digital Shadow that links the process data of recycled carbon fibre tapes to their measured quality. My studies centre on damage and fracture mechanics; alongside them, I work on Ansys projects in composite design and stress analysis, each validated with analytical methods. Previously, two years of engineering experience in India. Seeking an entry-level position in structural simulation, composites or materials engineering.

What I bring to your team

- ▶ **Validated structural simulation:** Ansys Mechanical, mesh convergence and boundary-condition checks, every result compared with analytical theory.
- ▶ **Composite analysis:** layered shells in Ansys ACP, laminate theory by hand and in software, Tsai-Wu, Puck and Hashin failure.
- ▶ **Damage and fracture mechanics:** major subjects at RWTH Aachen, with fracture mechanics lab experience.
- ▶ **Process data and digitalisation:** Python pipelines turning process, laser and camera data into traceable datasets.
- ▶ **Flow simulation:** two-phase CFD in OpenFOAM, validated against laboratory flume experiments.

Experience

Fraunhofer Institute for Production Technology IPT

Master's thesis researcher

Oct 2025 – present

Aachen, Germany

- ▶ Developing a segment-based Digital Shadow that maps the condition of recycled UD carbon fibre tape along its length.
- ▶ Integrating process parameters, laser-based geometry and quality data and camera triggers into one traceable Python dataset.
- ▶ Designed a hierarchical ID schema that traces each tape segment from recycling parameters to measured quality.

Sutherland Global Services, *technical associate (McAfee support), Kerala, India*

Resolved technical support cases; strengthened structured troubleshooting and customer communication.

Jun 2020 – Jan 2021

HomeScape Builders and Designers, *design engineer, Kerala, India*

Supported design development and execution with CAD, technical documentation and client coordination.

Jan 2020 – Jun 2020

SJC Marine Pvt. Ltd., *mechanical engineer, Kerala, India*

Supported mechanical design, maintenance and documentation in a marine engineering environment.

Jan 2019 – Jan 2020

Selected projects

700 bar hydrogen tank: sizing and failure analysis of the carbon fibre wall

2026

[Ansys ACP](#) [Ansys Mechanical](#) [Netting theory](#)

- Sized a type 4 tank wall (24 helical $\pm 15^\circ$ and 41 hoop plies, 16.25 mm) with netting theory for a burst ratio of 2.25.
- Layered-shell model with symmetry conditions; radial growth and fibre stresses within 1 % of classical laminate theory.
- Identified matrix cracking in the innermost helical plies (Puck) and a fibre-failure burst pressure of about 163 MPa, meeting the requirement.

First-ply failure of an open-hole carbon fibre laminate

2026

[Ansys ACP](#) [Ansys Mechanical](#) [Laminate theory](#)

- Modelled a quasi-isotropic $[0/45/-45/90]_s$ laminate as layered shells; laminate stiffness of 47.0 GPa matched classical laminate theory within 0.1 %.
- Predicted first-ply failure at 10.1 kN (Tsai-Wu) in the 90° plies at the hole edge, confirmed by hand calculation within 10 %.
- Found, explained and corrected an 88 mm spurious bending caused by a laminate offset, using an analytical cross-check.

Free-surface flow over a bed ridge (team of three)

2023

[OpenFOAM](#) [FreeCAD](#)

Univ. of Oldenburg / HS Emden-Leer

- Simulated two-phase open-channel flow with a locally refined mesh (blockMesh, snappyHexMesh), run in parallel on four cores.
- Validated against flume experiments: mean water-level deviation of about 2 cm (7 %).

Education

M.Sc. Materials Engineering, specialisation Structural Integrity

Apr 2023 – Mar 2027 (expected)

RWTH Aachen University

Aachen, Germany

Major subjects: damage mechanics, fracture mechanics, advanced solving methods in metal forming. Master's thesis at Fraunhofer IPT.

M.Sc. Engineering Physics

Apr 2022 – Mar 2023

Carl von Ossietzky University of Oldenburg

Oldenburg, Germany

Advanced fluid dynamics, OpenFOAM, Python for energy and meteorology analysis. Transferred to RWTH Aachen to specialise in materials and structural engineering.

Master's studies in Engineering Management

Nov 2021 – Apr 2022

IU International University of Applied Sciences (IUBH)

Germany

Transferred to a public university to pursue a technical Master's degree.

B.Tech Mechanical Engineering

2014 – 2018

TKM Institute of Technology (Cochin University of Science and Technology)

Kerala, India

Bachelor's project: pollution-sensing quadcopter with an embedded real-time sensor system. Best Food Technology Award (2018) for a centrifugal fat extraction concept for dairy waste treatment.

Skills and languages

Simulation

Ansys Mechanical, Ansys ACP, Ansys Discovery, Abaqus (coursework), OpenFOAM

Methods

Classical laminate theory, stress concentration analysis, failure criteria, mesh convergence, fracture mechanics

Programming

Python (data processing, data pipelines), C/C++

CAD

Creo, AutoCAD, FreeCAD, Ansys Discovery

Languages

English: fluent (C1) German: basic (A1/A2), actively improving

References available on request.